

Banking Management System

Software Requirements Specification (SRS)

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1. Introduction

1.1 Purpose

The Banking Management System (BMS) is a web-based application designed to automate and manage banking operations digitally. The system allows customers to create accounts, deposit and withdraw money, transfer funds, apply for loans, view transaction history, and manage profiles. Bank employees and administrators can manage customers, accounts, loans, transactions, and reports.

The objective is to provide secure, efficient, and reliable banking services while reducing manual workload and improving customer experience.

1.2 Scope

The Banking Management System provides:

- Customer account management
- Savings and current account operations
- Deposit and withdrawal management
- Fund transfer services
- Loan management
- Transaction tracking
- Online banking dashboard
- Employee management
- Branch management
- Reporting and analytics
- Authentication and authorization

1.3 Intended Audience

- Bank Customers
- Bank Employees
- Bank Managers
- System Administrators
- Software Developers
- Database Administrators

1.4 Definitions

Term	Description
Customer	Bank account holder
Account	Bank account maintained by customer
Transaction	Financial operation

Loan	Borrowed amount approved by bank
Admin	System administrator
Branch	Physical bank office

2. Overall Description

2.1 Product Perspective

The Banking Management System is a full-stack web application:

Layer	Technology
Frontend	React.js
Backend	Node.js · Express.js
Database (SQL)	MySQL
Database (NoSQL)	MongoDB
Version Control	GitHub
API Style	REST API

2.2 Product Functions

- Register customers
- Create bank accounts
- Perform deposits
- Perform withdrawals
- Transfer money
- Generate statements
- Manage loans
- Track transactions
- Manage employees
- Manage branches
- Produce reports

2.3 User Classes

Role	Capabilities
Customer	Register · Login · View balance · Deposit · Withdraw · Transfer · Apply loan · View statements
Employee	Verify customers · Approve accounts · Manage transactions · Process loans
Manager	Approve loans · View reports · Manage employees
Administrator	Manage system · Manage users · Configure branches · Generate analytics

3. Functional Requirements

Module 1: User Authentication

ID	Requirement
FR-1	System shall allow users to register.
FR-2	System shall allow users to login.
FR-3	System shall support password encryption.
FR-4	System shall support JWT authentication.

Module 2: Customer Management

ID	Requirement
FR-5	Create customer profile.
FR-6	Update customer information.
FR-7	Delete customer account.
FR-8	View customer profile.

Information Fields:

- Customer ID
- Full Name
- Date of Birth
- Gender
- Email
- Phone
- Address
- National ID

Module 3: Bank Account Management

ID	Requirement
FR-9	Create savings account.
FR-10	Create current account.
FR-11	Close account.
FR-12	View account details.

Information Fields:

- Account Number

- Account Type
- Balance
- Branch
- Status

Module 4: Deposit Management

ID	Requirement
FR-13	Deposit money.
FR-14	Generate deposit receipt.
FR-15	Update balance automatically.

Module 5: Withdrawal Management

ID	Requirement
FR-16	Withdraw money.
FR-17	Validate available balance.
FR-18	Generate withdrawal receipt.

Module 6: Fund Transfer

ID	Requirement
FR-19	Transfer funds between accounts.
FR-20	Validate sender balance.
FR-21	Record transfer transaction.
FR-22	Send transfer notification.

Module 7: Transaction Management

ID	Requirement
FR-23	Record all transactions.
FR-24	Generate transaction history.
FR-25	Filter by date.
FR-26	Generate mini statement.

Transaction Types:

- Deposit

- Withdrawal
- Transfer
- Loan Payment

Module 8: Loan Management

ID	Requirement
FR-27	Submit loan application.
FR-28	Review loan request.
FR-29	Approve loan.
FR-30	Reject loan.
FR-31	Calculate EMI.

Loan Types:

- Personal Loan
- Education Loan
- Home Loan
- Business Loan

Module 9: Employee Management

ID	Requirement
FR-32	Add employee.
FR-33	Update employee.
FR-34	Delete employee.
FR-35	Assign branch.

Module 10: Branch Management

ID	Requirement
FR-36	Add branch.
FR-37	Update branch.
FR-38	Delete branch.
FR-39	Assign manager.

Module 11: Reports

ID	Requirement
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FR-40	Generate customer report.
FR-41	Generate transaction report.
FR-42	Generate loan report.
FR-43	Generate branch performance report.
FR-44	Generate monthly financial report.

4. Non-Functional Requirements

Category	Requirement
Performance	Support 1000+ concurrent users; response time under 3 seconds; DB queries optimised with indexes.
Reliability	99.9% uptime; daily automated backup.
Security	Password hashing via bcrypt; JWT authentication; role-based access control; HTTPS; SQL injection protection.
Scalability	Modular architecture; horizontal scaling support.
Maintainability	RESTful architecture; MVC pattern; clean code principles.
Availability	System available 24/7.

5. Use Cases

UC-01 — Customer Registration

Actor: Customer

1. Open registration page.
2. Enter details.
3. Submit form.
4. System validates data.
5. Account created.

UC-02 — Deposit Money

Actor: Customer

1. Login.
2. Select account.
3. Enter amount.
4. Confirm deposit.
5. Balance updated.

UC-03 — Withdraw Money

Actor: Customer

1. Login.
2. Enter withdrawal amount.
3. Validate balance.
4. Complete transaction.

UC-04 — Transfer Money

Actor: Customer

1. Login.
2. Select sender account.
3. Enter receiver account.
4. Enter amount.
5. Confirm transfer.
6. Update balances.

UC-05 — Apply for Loan

Actor: Customer

1. Fill application.
2. Upload documents.
3. Submit request.
4. Employee reviews.
5. Manager approves/rejects.

6. Database Requirements

Main Entities

#	Entity
1	Users
2	Customers
3	Accounts
4	Transactions
5	Loans
6	Employees
7	Branches
8	Roles

Entity Relationships

Relationship	Cardinality
Customer → Account	One-to-Many (1:M)
Account → Transaction	One-to-Many (1:M)
Customer → Loan	One-to-Many (1:M)
Branch → Employee	One-to-Many (1:M)
Branch → Account	One-to-Many (1:M)

7. System Architecture

Layer	Technology	Role
Presentation	React.js	User interface / single-page application
Communication	REST API	HTTP request/response (JSON)
Application	Node.js / Express.js	Business logic, routing, middleware
Primary DB	MySQL	Structured data: accounts, transactions, loans
Secondary DB	MongoDB	Logs, notifications, audit trail

8. Future Enhancements

- Mobile banking application
- QR code payments
- ATM integration
- Credit card management
- AI-powered fraud detection
- SMS notifications
- Online cheque processing

9. Conclusion

The Banking Management System will provide a secure and efficient platform for managing banking operations digitally. The project integrates Software Engineering, Database Theory, and Database Lab concepts while demonstrating full-stack development skills using React, Node.js, MySQL, and MongoDB.

By implementing role-based access control, JWT-secured endpoints, and a dual-database architecture, the BMS meets modern standards for reliability, scalability, and maintainability — making it a suitable foundation for a real-world banking platform.